These theorems are :

(*a*) Superposition theorem

(*b*) Thevenin theorem

(*c*) Norton theorem

(*d*) Maximum power transfer theorem

First three theorems are used to find out the response while the fourth theorem is used to determine the value of load resistance for which source transfers the maximum power.

3.3.1. Super Position Theorem. Superposition theorem states that in any linear, active, bilateral network, having more than one source, response across any element is the sum of responses obtained by each source considered separately and all other sources are replaced by their internal resistances.

Explanation. Consider the circuit shown in Fig. 3.6, where two sources (voltage and current sources) are acting together.

If we want to determine current through resistance, R_3, then to apply superposition theorem we take one source at a time.

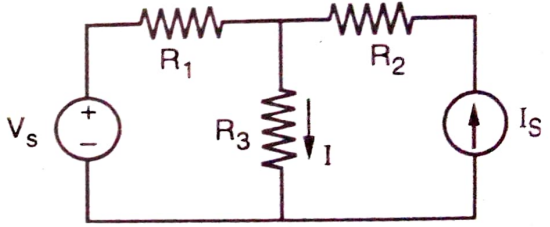


Fig. 3.6.

(*i*) **Case I.** Take the voltage source and current source is open circuited.

(Since the internal resistance of ideal current source is infinite, so it is open circuited).

$$I_1 = \frac{V_s}{R_1 + R_3} \qquad ...(3.19)$$

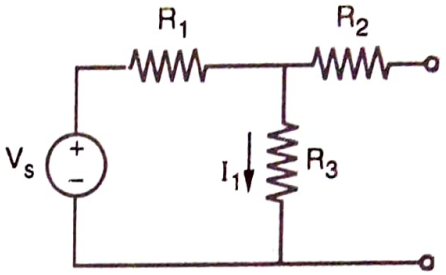


Fig. 3.7.

(*ii*) **Case II.** Now take the current source and short the voltage source.

(Since the internal resistance of ideal voltage source is zero, so it is short circuited).

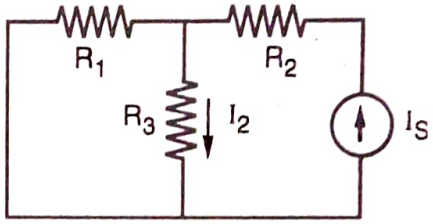


Fig. 3.8.

By current division rule :

$$I_2 = I_s \times \frac{R_1}{R_1 + R_3} \qquad ...(3.20)$$

Now if both the sources acting together, then by superposition theorem

response $\qquad I = I_1 + I_2 \qquad ...(3.21)$

Super position principle is combination of two properties :

(*i*) Additivity

(*ii*) Homogenity.

(*i*) **Additivity.** If network has y_1 output for x_1 input and output y_2 for input x_2 then it must has $y_1 + y_2$ output for input $x_1 + x_2$. This property of circuit is called additivity.

(*ii*) **Homogenity.** If input is multiplied by any constant K then output must be multiplied by same constant. This property is called homogenity.

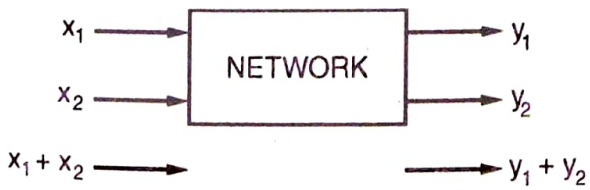


Fig. 3.9. Additivity

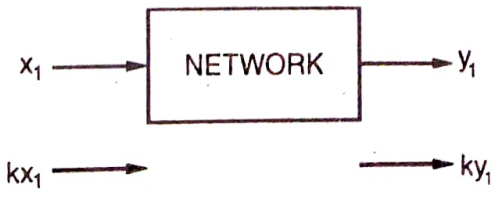


Fig. 3.10. Homogenity.

Limitations of superposition theorem :

(*i*) This theorem is strictly applicable to linear networks only.

(*ii*) Power cannot be determined directly by this theorem because power is not linearly related with source voltage or current.

(*iii*) Dependent sources are not suppressed if they are externally controlled.

 Thev... This theorem was proposed to solve the complicated

EXAMPLES ON SUPERPOSITION THEOREM :

Example 3.25. *Find current-I by superposition theorem.*

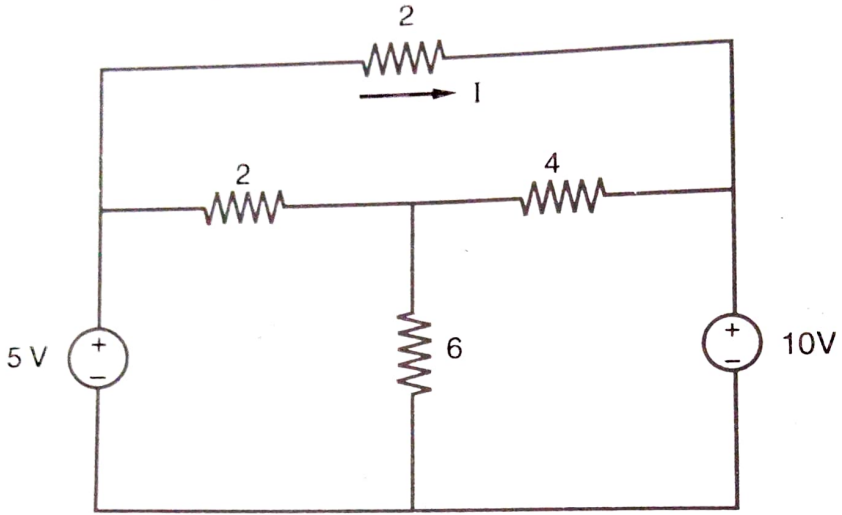


Fig. 3.62.

Solution. First we take the 5 V source and 10 V source is replaced by its internal resistance (since source is ideal so it is short-circuited).

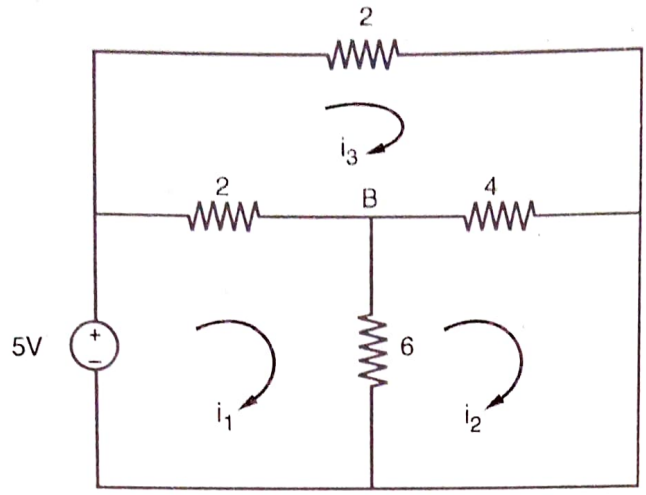


Fig. 3.63.

Here we have to find current i_3. We apply the Maxwell loop method.

In loop 1 : $2(i_1 - i_3) + 6(i_1 - i_2) = 5$

$\Rightarrow$ $8\,i_1 - 6\,i_2 - 2\,i_3 = 5$ ...(*i*)

In loop 2 : $6(i_2 - i_1) + 4(i_2 - i_3) = 0$

$\Rightarrow$ $-6\,i_1 + 10\,i_2 - 4\,i_3 = 0$ ...(*ii*)

In loop 3 : $2(i_3 - i_1) + 2\,i_3 + 4(i_3 - i_2) = 0$

$\Rightarrow$ $-2\,i_1 - 4\,i_2 + 8\,i_3 = 0$ ...(*iii*)

Since we are interested in current i_3 only so we apply the Cramer's rule to find it,

The equations (*i*), (*ii*) and (*iii*) can be written in matrix form :

$$\begin{bmatrix} 8 & -6 & -2 \\ -6 & 10 & -4 \\ -2 & -4 & 8 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$$

Now i_3 is given by (using Cramer's rule).

$$i_3 = \frac{\begin{vmatrix} 8 & -6 & 5 \\ -6 & 10 & 0 \\ -2 & -4 & 0 \end{vmatrix}}{\begin{vmatrix} 8 & -6 & -2 \\ -6 & 10 & -4 \\ -2 & 4 & 8 \end{vmatrix}}$$

$$= \frac{5\,(24 + 20)}{8\,(80 - 16) + 6\,(-48 - 8) - 2\,(24 + 20)}$$

$$= \frac{220}{88} = 2.5 \text{ Amp.}$$

Now we take the 10 V source in consideration and short circuited the 5 V source (Fig. 3.64).

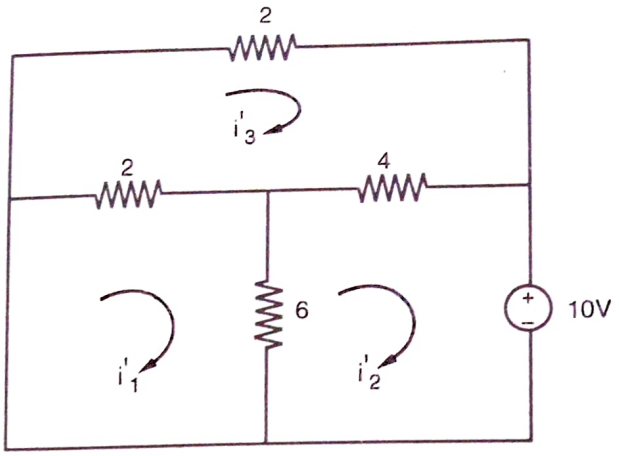


Fig. 3.64.

Now we have to calculate the current i'_3. So by loop method we have the following equations :

In loop 1 : $2(i'_1 - i'_3) + 6(i'_1 - i'_2) = 0$

$\Rightarrow$ $8i_1' - 6i_2' - 2i_3' = 0$ ...(i)

In loop 2 : $6(i'_2 - i'_1) + 4(i'_2 - i'_3) = -10$

$\Rightarrow$ $-6i'_1 + 10i'_2 - 4i'_3 = -10$ ...(ii)

In loop 3 : $2(i'_3 - i'_1) + 4(i'_3 - i'_2) + 2i'_3 = 0$

$\Rightarrow$ $-2i'_1 - 4i'_2 + 8i'_3 = 0$ ...(iii)

Equations (*i*), (*ii*) and (*iii*) in matrix form :

$$\begin{bmatrix} 8 & -6 & -2 \\ -6 & 10 & -4 \\ -2 & -4 & 8 \end{bmatrix} \begin{bmatrix} i'_1 \\ i'_2 \\ i'_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -10 \\ 0 \end{bmatrix}$$

Now i'_3 can be calculated by using the Cramer's rule.

$$i'_3 = \frac{\begin{vmatrix} 8 & -6 & 0 \\ -6 & 10 & -10 \\ -2 & -4 & 0 \end{vmatrix}}{\begin{vmatrix} 8 & -6 & -2 \\ -6 & 10 & -4 \\ -2 & -4 & 8 \end{vmatrix}} = \frac{-440}{88} = -5 \text{ Amp.}$$

So the resultant current-I due to both the sources (5 V and 10 V) is the summation of i_3 and i'_3 by superposition theorem

$$I = i_3 + i'_3 = 2.5 - 5 = -2.5 \text{ Amp.}$$

Example 3.26. *Find power loss in 4 Ω resistor. (Fig. 3.65)*

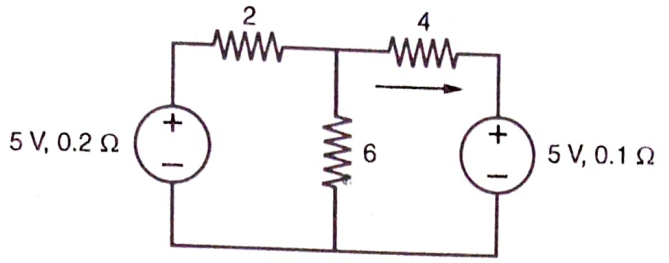


Fig. 3.65.

Solution. First we find the current through 4 Ω resistor by super position theorem.

Here both the sources have the internal resistances. Take one voltage source and replace the other by its internal resistance.

Current passing through 4 Ω resistor = i_2. (Fig. 3.66)

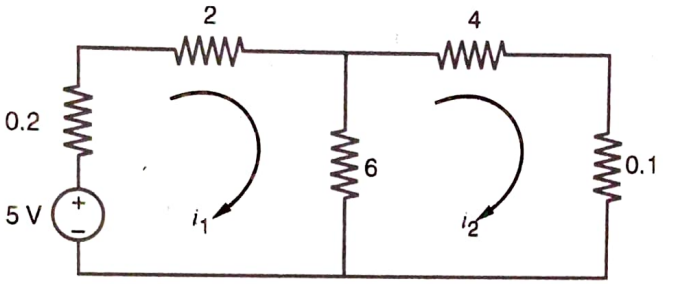


Fig. 3.66.

By loop method :

In loop 1 : $8.2\, i_1 - 6\, i_2 = 5$ ...(*i*)

In loop 2 : $-6\, i_1 + 10.1\, i_2 = 0$ ...(*ii*)

After solving equations (*i*) and (*ii*)

We get $i_2 = 0.64$ Amp.

Now we take the another source (5 *V*, 0.1 Ω), then the current through 4 Ω resistor is i'_2 as shown in Fig. 3.67.

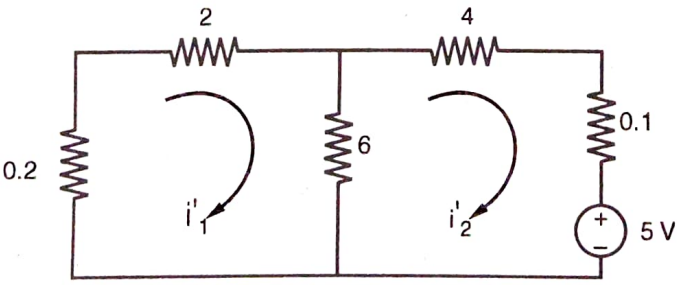


Fig. 3.67.

We again apply the loop method :

In loop 1 : $8.2\, i'_1 - 6\, i'_2 = 0$ ...(*i*)

In loop 2 : $-6\, i'_1 + 10.1\, i'_2 = -5$ ...(*ii*)

After solving the equations (*i*) and (*ii*) we get,

$$i'_2 = -0.875 \text{ Amp.}$$

So total current passing through 4 Ω resistor by superposition theorem is

$$i = i_2 + i'_2 = 0.64 - 0.875 = -0.235 \text{ Amp.}$$

Power loss in 4 Ω resistor $P = i^2 R = (0.235)^2 \times 4 = 0.221$ watts.

Example 3.27. *Find current through 6 Ω resistor. (Fig. 3.68.)*

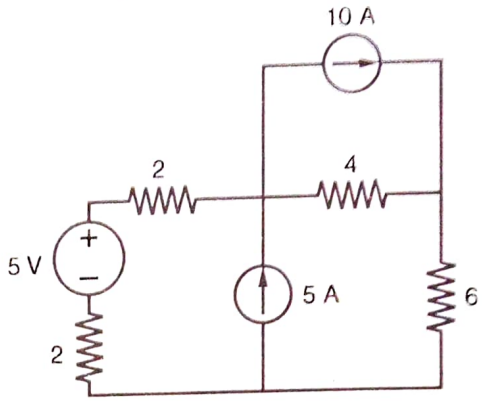


Fig. 3.68.

Solution. **Case I.** 5 A current source is taken. (Fig. 3.69).

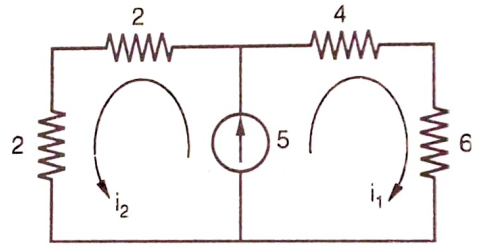


Fig. 3.69.

By current division rule : $i_1 = 5 \times \frac{4}{14} = 1.428$ Amp.

Case II. Current source of 10 A is taken into consideration.

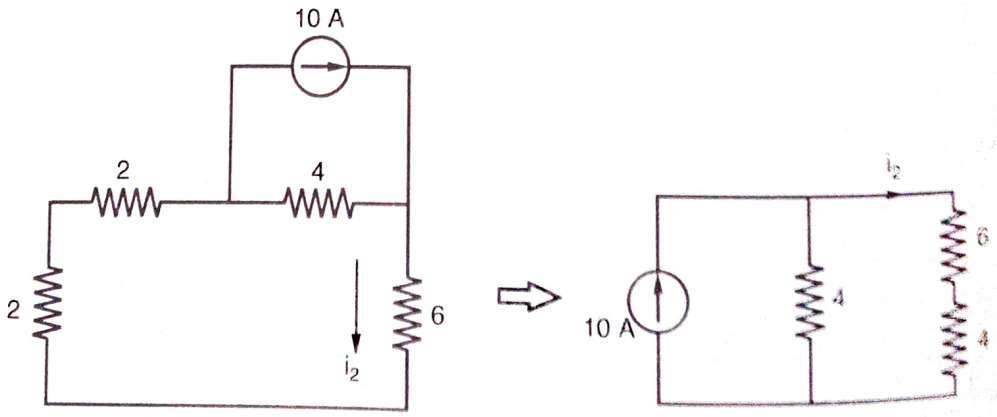


Fig. 3.70.

By current division rule : $i_2 = 10 \times \frac{4}{14} = 2.857$ Amp.

Case III. 5 V source is taken into consideration.

$$i_3 = \frac{5}{14} = 0.357 \text{ Amp.}$$

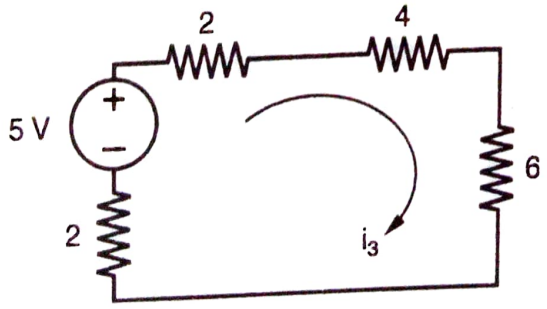


Fig. 3.71.

So by superposition current through 6 Ω resistor

$$i = i_1 + i_2 + i_3 = 1.428 + 2.857 + 0.357 = 4.642 \text{ Amp.}$$

Limitations of superposition theorem :

(*i*) This theorem is strictly applicable to linear networks only.

(*ii*) Power cannot be determined directly by this theorem because power is not linearly related with source voltage or current.

(*iii*) Dependent sources are not suppressed if they are externally controlled.

3.3.2. Thevenin Theorem. This theorem was proposed to solve the complicated networks in 1883 by the french telegraph engineer Leon Thevenin. Then Helmholtz gave the present shape to the work of Thevenin in the form of following statement.

Any linear, active, bilateral complicated network across its load terminals can be replaced by single voltage source and one series resistance.

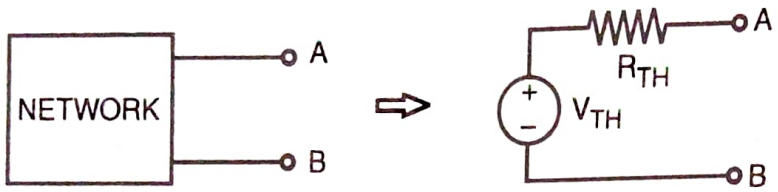


Fig. 3.11.

where V_{TH} = Thevenin's equivalent voltage

= It is the open circuit voltage across terminals AB (load terminals)

R_{TH} = Thevenin's equivalent resistance as seen from the load terminals when all the sources replaced by their internal impedances.

= It is the driving point impedance at load terminals AB.

Procedures for Thevenin theorem :

(*i*) First of all remove the load resistance, if any (load is that element where the response is desired).

(*ii*) Now determine the open circuit voltage across the load terminals by using Maxwell loop or node method. This is V_{TH}.

(*iii*) Replace all the independent sources by their internal resistance. (If sources are ideal then short the voltage source and open the current source). Now find the equivalent resistance at the load terminals. This is R_{TH}.

(*iv*) Now draw the Thevenin's equivalent circuit as shown in Fig. 3.12. Then connect the load resistance (if any) and determine the response as desired.

Load current $$I_L = \frac{V_{TH}}{R_{TH} + R_L} \qquad ...(3.22)$$

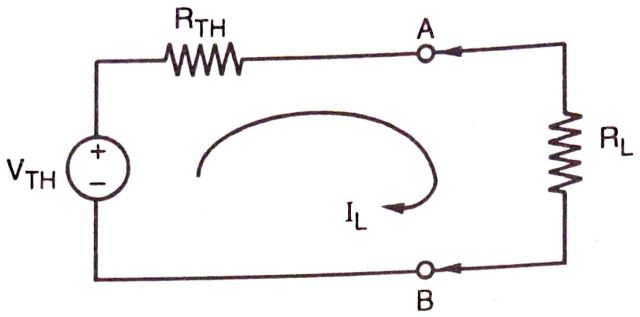


Fig. 3.12. Thevenin's Equivalent Circuit.

Advantages :

(*i*) Thevenin theorem is used to convert the multisource, multiloop and multinode networks to simple network having one source and one resistance.

(*ii*) If there is a single non linear element in any circuit then it can be considered as load and removed from the circuit.

Now the rest circuit can be converted into Thevenin equivalent circuit. In this way Thevenin theorem separates the linear and non linear elements.

(*iii*) Before applying the maximum power transfer theorem, circuit is first converted into Thevenins equivalent circuit.

Example 3.28. *Find power loss in 3 Ω resistor by Thevenin theorem.*

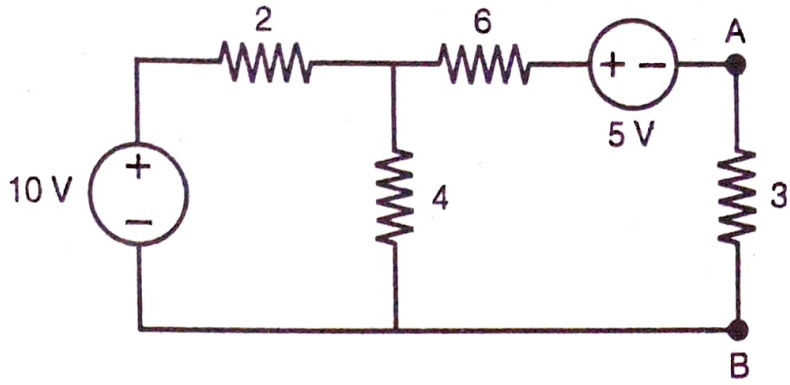


Fig. 3.72.

Solution. Remove the load $R_L = 3\ \Omega$.

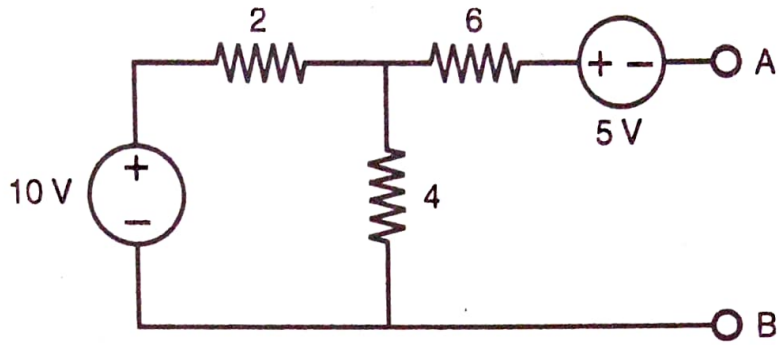


Fig. 3.73.

Calculation for R_{TH} :

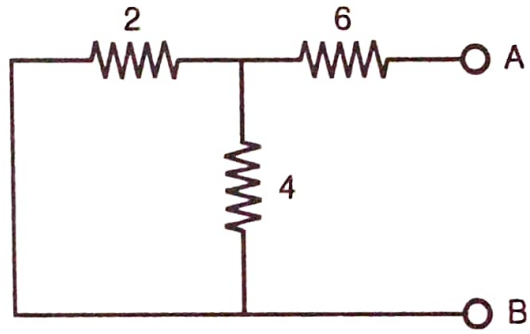


Fig. 3.74.

$$R_{TH} = 2 \parallel 4 + 6 = \frac{2 \times 4}{2+4} + 6 = 7.33\ \Omega$$

Calculation for V_{TH} :

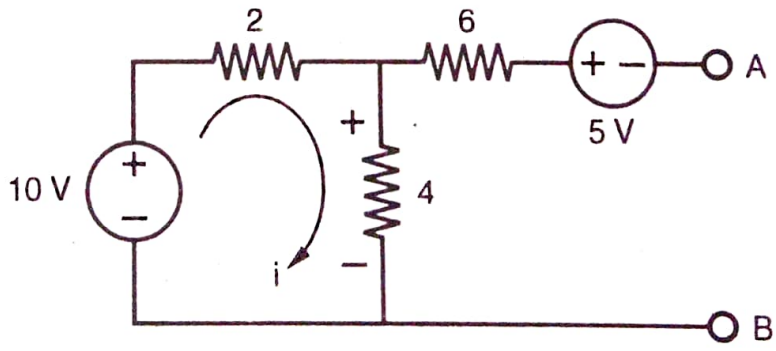


Fig. 3.75.

$$i = \frac{10}{2+4} = \frac{10}{6} \text{ Amp.}$$

$$V_{oc} = -5V + 4\left(\frac{10}{6}\right) = 1.66 \text{ Volt}$$

Thevenin's equivalent circuit.

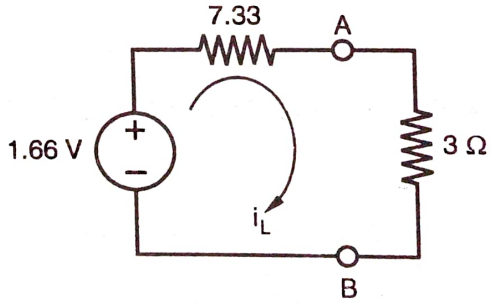


Fig. 3.76.

$$i_L = \frac{1.66}{7.33+3} = 0.16 \text{ Amp}$$

Power loss $P_L = i_L^2 \times 3 = 0.078$ Watts.

3.3.3. Norton's Theorem. The Dual of thevenin theorem was given by E.L. Norton of the Bell Telephone Laboratories. According to Norton theorem :

"Any linear, active, bilateral complicated network across its load terminals can be replaced by single current source and one parallel resistance".

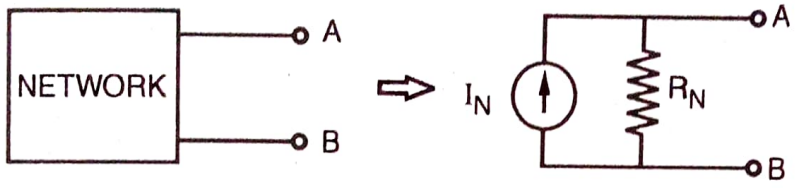


Fig. 3.13.

where, I_N = Norton's equivalent current

= It is the short circuit current passing through terminals AB, when they are short circuited.

and R_N = Norton's equivalent resistance

= R_{TH} (equivalent resistance at terminals AB, when all independent sources replaced by their internal resistances)

Procedure for Norton theorem :

(*i*) First of all remove the load, if any.

(*ii*) Short circuit the load terminals AB and determine the short circuit current passing through it. This is I_N.

(*iii*) Now replace all the independent sources by their internal resistance and find the equivalent resistance as seen from load terminals AB. This is R_N and equal to R_{TH} .

(*iv*) Now draw the Norton's equivalent circuit. Connect the load and determine the response as desired.

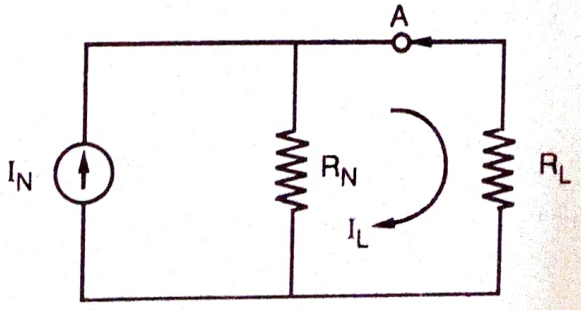


Fig. 3.14. Nortons's Equivalent Circuit.

By current division rule, the current through the load

$$I_L = I_N \times \frac{R_N}{R_N + R_L} \qquad ...(3.23)$$

Example 3.29. *Solve the previous problem by Norton theorem.*

Solution. The Norton's equivalent resistance

$$R_N = R_{TH} = 7.33\ \Omega$$

Norton's equivalent current. We short the load terminals.

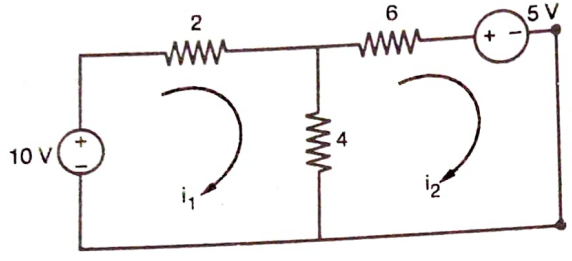


Fig. 3.77.

Now we calculate the current passing through the short circuited load terminals by loop method. This current will be the Norton's equivalent current I_N. In Fig. 3.77 i_2 i the I_N

In loop 1 : $2\,i_1 + 4\,(i_1 - i_2) = 10$

$\Rightarrow$ $\qquad 6\,i_1 - 4\,i_2 = 10$ ...(

In loop 2 : $4\,(i_2 - i_1) + 6\,i_2 = -5$

$\Rightarrow$ $\qquad -4\,i_1 + 10\,i_2 = -5.$ ...(

After solving these equations, we get,

$$I_N = i_2 = 0.227 \text{ Amp.}$$

Now we draw the Norton's equivalent circuit.

Norton equivalent circuit.

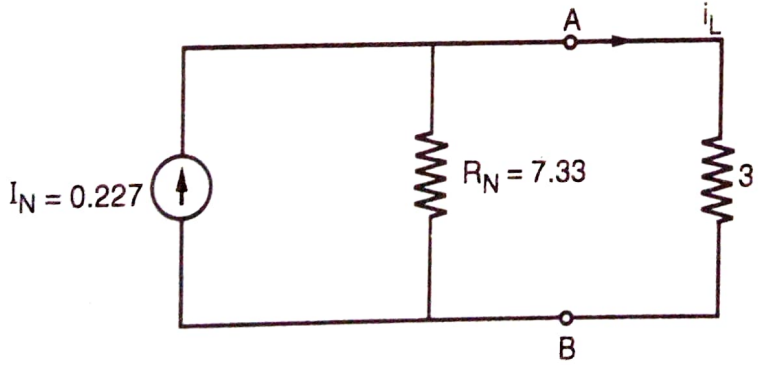


Fig. 3.78.

By current division rule : $i_1 = i_N \times \dfrac{R_N}{R_N + 3} = 0.227 \times \dfrac{7.33}{7.33 + 3} = 0.16$ Amp.

Power loss $P_L = i_L^2 \times 3 = (0.16)^2 \times 3 = 0.078$ watts.

Example 3.33. *Find I by Thevenin theorem :*

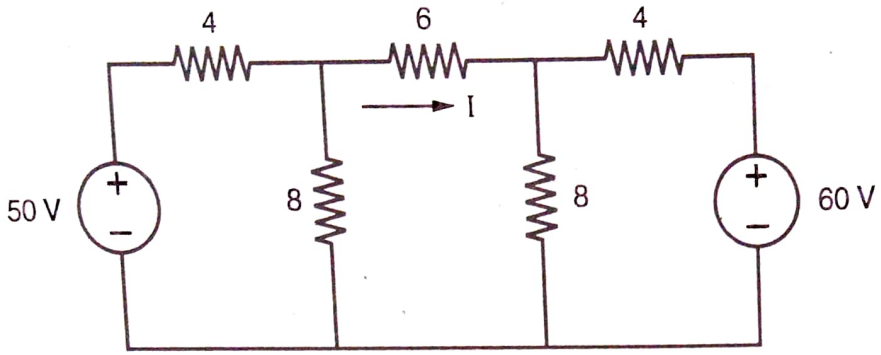


Fig. 3.85.

Solution. Calculation for R_{TH} : Short the voltage sources.

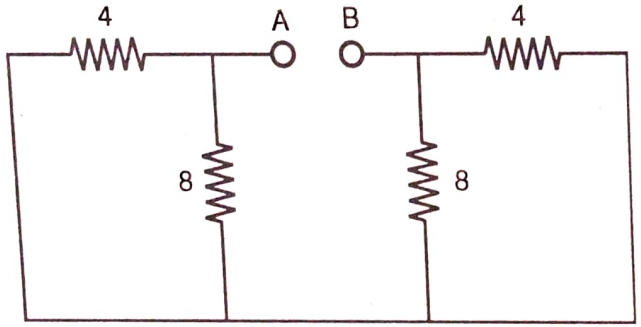


Fig. 3.86.

Now 4 Ω and 8 Ω resistances are in parallel

$$R_{TH} = 4 \parallel 8 + 8 \parallel 4 = \frac{4 \times 8}{8+4} + \frac{8 \times 4}{8+4} = 5.33\ \Omega.$$

Calculation for V_{TH} :

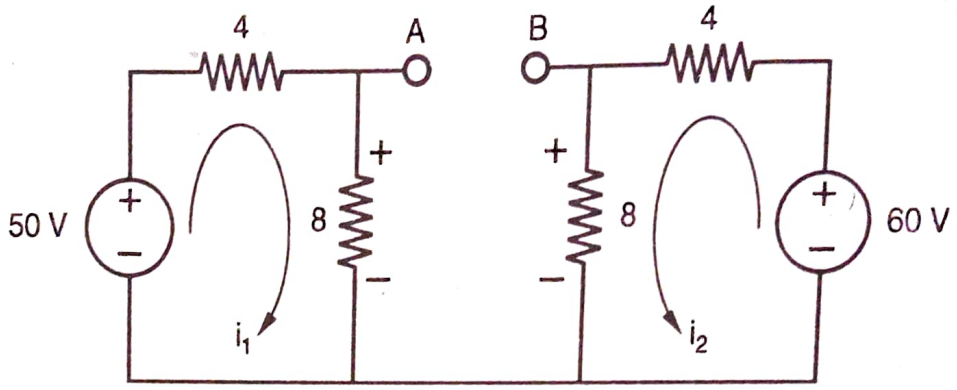


Fig. 3.87.

$$i_1 = \frac{50}{12} = 4.16 \text{ Amp.}$$

$$i_2 = \frac{60}{12} = 5 \text{ Amp.}$$

$$V_{TH} = V_{AB} = V_A - V_B = 8\, i_1 - 8 i_2 = 8 \times 4.16 - 8 \times 5 = -6.67 \text{ Volt}$$

Thevenin's equivalent circuit :

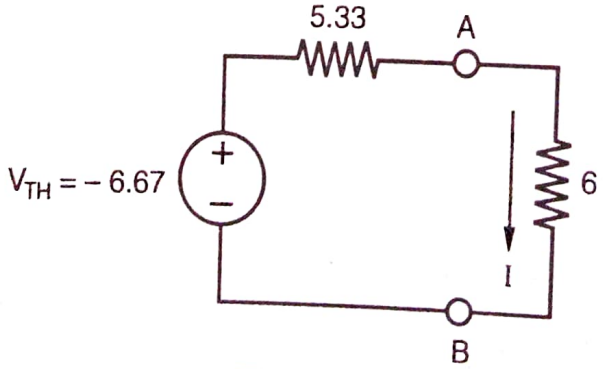


Fig. 3.88.

$$I = -\frac{6.67}{5.33 + 6} = -0.589 \text{ Amp.}$$

Maximum power

Example 3.35. *Solve the example 3.31 by Norton's theorem.*

Solution. $R_N = R_{TH} = 5.33\ \Omega$

Calculation for I_N : Short the load terminals

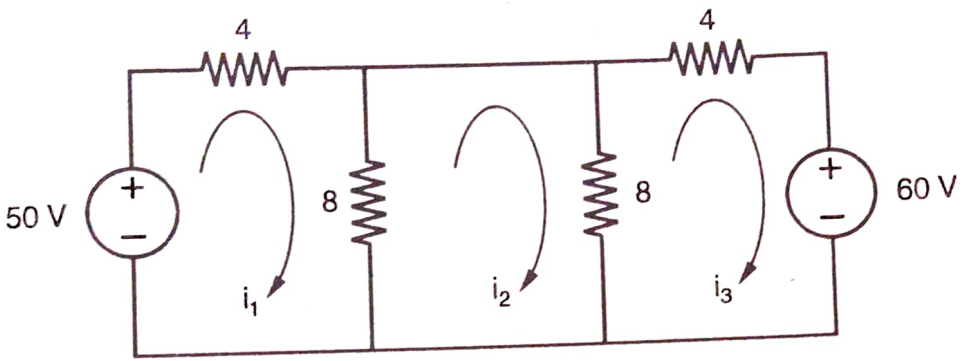


Fig. 3.91.

$$I_N = i_2 = \frac{\begin{vmatrix} 12 & 50 & 0 \\ -8 & 0 & 8 \\ 0 & 60 & 12 \end{vmatrix}}{\begin{vmatrix} 12 & -8 & 0 \\ -8 & 16 & 8 \\ 0 & 8 & 12 \end{vmatrix}} = -\frac{960}{768} = -1.25 \text{ Amp.}$$

Norton's equivalent circuit : (Fig. 3.92)

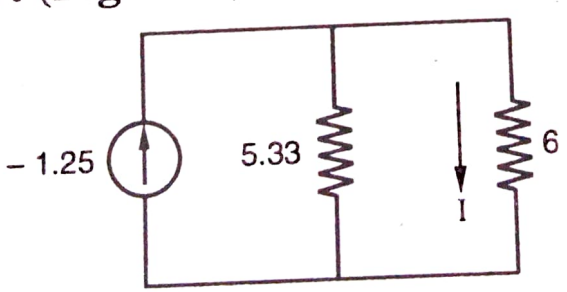


Fig. 3.92.

$$I = -1.25 \times \frac{5.33}{6 + 5.33} = -0.589 \text{ Amp.}$$

Example 3.36. *Find the power in 8 Ω resistor, connected across AB by Thevenin theorem.*

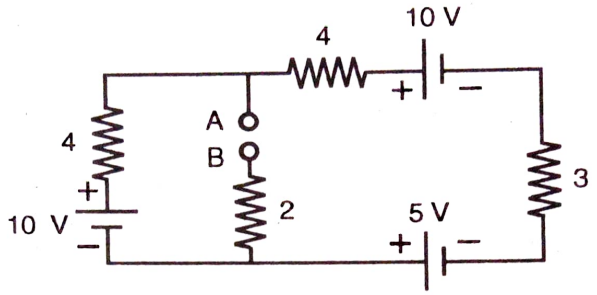


Fig. 3.93.

Solution. Calculation for R_{TH} : Short all the three voltage sources.

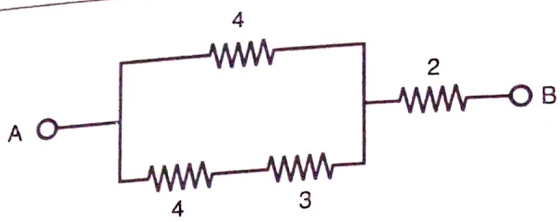


Fig. 3.94.

$$R_{TH} = \frac{4 \times 7}{4+7} + 2 = 4.545\ \Omega.$$

Calculation for V_{TH} :

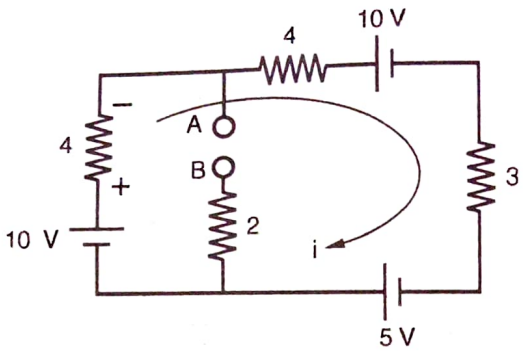


Fig. 3.95.

$$i = \frac{10 - 10 + 5}{4 + 4 + 3} = \frac{5}{11} \text{ Amp.}$$

$$V_{TH} = 10 - 4i = 10 - 4 \times \frac{5}{11} = \frac{90}{11} = 8.182 \text{ Volt.}$$

Thevenin's equivalent circuit : (Fig. 3.96)

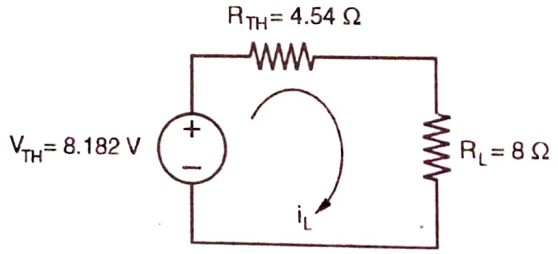


Fig. 3.96.

$$i_L = \frac{V_{TH}}{R_{TH} + R_L} = \frac{8.182}{4.54 + 8} = 0.652 \text{ Amp.}$$

Power loss in 8 Ω resistor.

$$P = (i_L)^2 \times 8 = 3.40 \text{ Watts.}$$

Example 3.37. *Solve the above problem by Norton's theorem.*

Solution. Norton's equivalent resistance

$$R_N = R_{TH} = 4.54\ \Omega.$$

Norton's equivalent current I_N :

First we short circuit the terminals A and B. Then we find the current passing through the short circuited terminals. Here we choose the loops as shown in Fig. 3.97. Now the Norton's equivalent current $I_N = i_1$

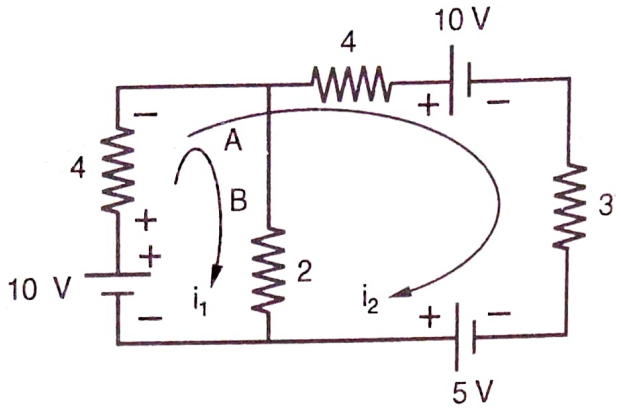


Fig. 3.97.

KVL in loop 1 : $4\,(i_1 + i_2) + 2\,i_1 = 10$

$\Rightarrow$ $6\,i_1 + 4\,i_2 = 10$ ...(*i*)

In loop 2 : $4\,(i_1 + i_2) + 4\,i_2 + 3\,i_2 = 10 - 10 + 5$

$\Rightarrow$ $4\,i_1 + 11\,i_2 = 5$ ...(*ii*)

From equations (*i*) and (*ii*) $I_N = i_1 = 1.8$ Amp.

Norton equivalent circuit : (Fig. 3.98)

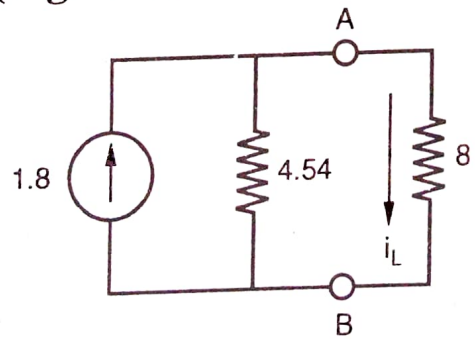


Fig. 3.98.

$$i_L = 1.8 \times \frac{4.54}{8 + 4.54} = 0.652 \text{ Amp.}$$

Power loss in 8 Ω $p = (i_L)^2 \times 8 = 3.40$ Watts.

3.3.4. Maximum Power transfer theorem. The maximum power transfer theorem states that *"In the Thevenin's equivalent circuit the source will transfer the maximum power to load resistance if the value of this load resistance is equal to the Thevenin equivalent resistance (or the internal resistance)."*

Fig. 3.15 shows the Thevenin's equivalent circuit of a network feeding power to a load resistance R_L.

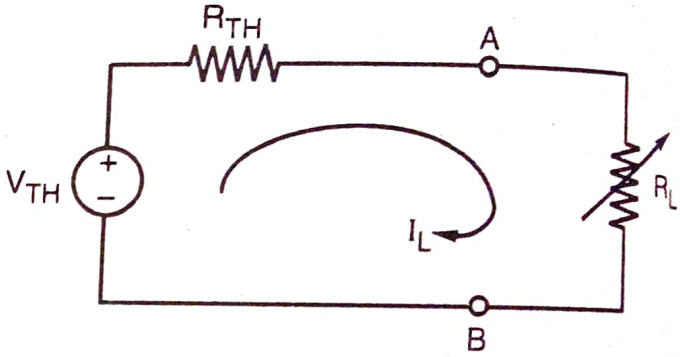


Fig. 3.15.

According to maximum power transfer theorem, the source V_{TH} will transfer the maximum power to load if

$$R_L = R_{TH}$$

Proof. The load current I_L is given by

$$I_L = \frac{V_{TH}}{R_{TH} + R_L} \qquad \ldots(3.26$$

The power drawn by load R_L is

$$P_L = I_L^2 R_L = \left(\frac{V_{TH}}{R_{TH} + R_L} \right)^2 R_L$$

For power to be maximum $\frac{dP_L}{dR_L} = 0$

$$\frac{d}{dR_L} \left[\frac{(V_{TH}^2)}{(R_{TH} + R_L)^2} R_L \right] = 0$$

$$(R_{TH} + R_L)^2 - R_L\, 2\,(R_{TH} + R_L) = 0$$

$$R_{TH} + R_L - 2R_L = 0$$

$$R_L = R_{TH}.$$

...(3.27)

Example 3.30. *What load resistance should be connected across the terminals AB to receive the maximum power. Also calculate the value of maximum power.*

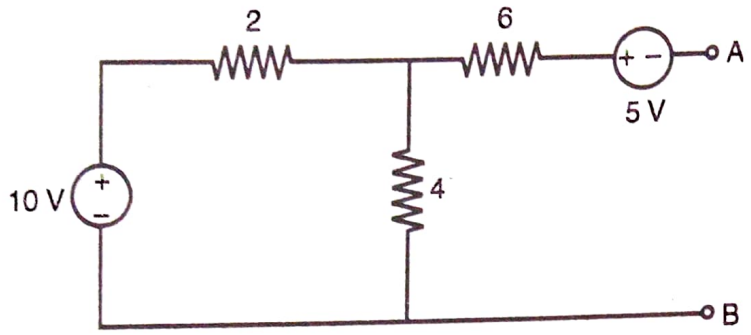


Fig. 3.79.

Solution. For applying the maximum power transfer theorem, first we have to draw the Thevenin's equivalent circuit, which we have already drawn in example 3.29.

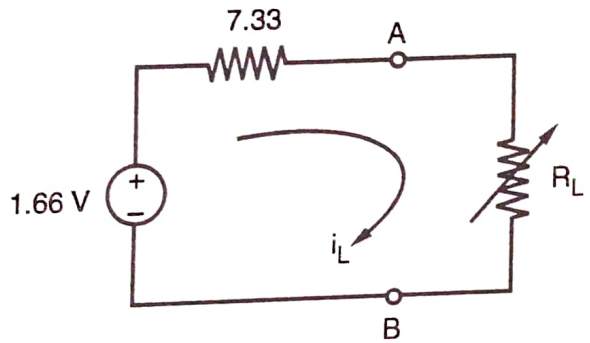


Fig. 3.80.

Now the value of load resistance R_L for receiving the maximum power, should be equal to the R_{TH} (according to maximum power transfer theorem).

$$R_L = R_{TH} = 7.33$$

$\therefore$ Load current $$i_L = \frac{1.66}{7.33 + 7.33} = 0.1132 \text{ Amp.}$$

So the maximum power delivered to the load

$$P_{max} = i_L^2 \times R_L = (0.1132)^2 \times 7.33 = 0.0939 \text{ Watts.}$$